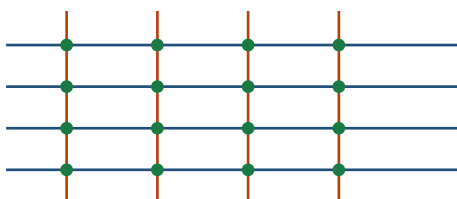


Block Size Is Tile Height

Storing 2-Bit Floating-Point Neural Networks
Inside a ReRAM Crossbar

*A complete explanation, starting from
how a single memory cell works*



Shaivi Nandi

International Institute of Information Technology, Bangalore

August 18, 2026

How to read this document

This document explains a research project from the ground up. It assumes you know what a resistor is and what a neural network does, and nothing else. Every piece of jargon is defined the first time it appears.

The story has one central idea, and if you remember only one thing, remember this:

The whole project in four sentences

We store neural network weights as physical resistances in a memory chip, so that multiplication happens through Ohm's law instead of in a digital multiplier. When we did this the obvious way, the network's accuracy collapsed from 92% to 10% — no better than random guessing. We found the cause was not random noise, as the published literature assumes, but a *predictable shrinking* of every output by a factor we can calculate in advance. Dividing that factor back out recovers the accuracy exactly, and because the shrinkage was the only thing limiting how large we could build the array, removing it also made the design 13.6× more energy efficient.

The structure:

- **Part I** explains the hardware: what a ReRAM cell is, how it stores a number, and how a grid of them multiplies.
- **Part II** explains the number format (FP2) and why it collides with the hardware. This collision is the problem the project solves.
- **Part III** derives the fix mathematically. This is the contribution.
- **Part IV** presents every experimental result, with the tables and the reasoning behind each.
- **Part V** is preparation for questions — what is genuinely new, what was retracted and why, and what is not yet done.

Boxes are colour-coded: **blue** for key ideas, **green** for “how to say this out loud”, and **orange** for warnings, limitations and honest caveats.

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Part I

The Hardware, From Scratch

Chapter 1

What a ReRAM Cell Is

1.1 The one-sentence version

A **ReRAM cell** (Resistive Random-Access Memory, also written RRAM) is a sandwich of two metal electrodes with a thin insulating oxide between them. Normally the oxide does not conduct. But if you apply a large enough voltage, a microscopic conducting path forms through it, and the sandwich becomes a conductor. Apply a large voltage the other way, and the path breaks.

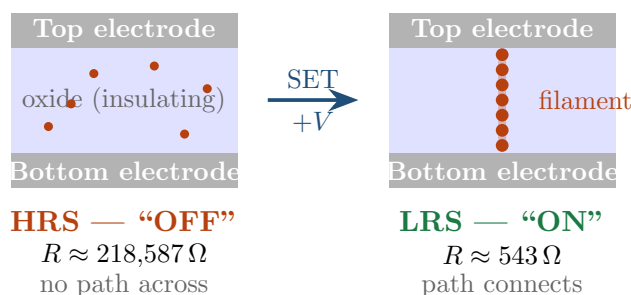


Figure 1.1: A ReRAM cell in its two extreme states. Oxygen vacancies (orange) scattered randomly leave the cell insulating (HRS). A voltage pulse lines them up into a conducting *filament*, and the cell becomes a good conductor (LRS). The resistance values shown are the ones measured in our SPICE calibration and used throughout this work.

1.2 Why this is useful

Three properties matter for us.

It is non-volatile. The filament stays where it is when power is removed. A ReRAM cell remembers its value with no electricity at all. A conventional SRAM cell needs six transistors constantly powered to hold one bit; the moment you cut power, the bit is gone. This is why our standby power comparison later shows 0 mW against SRAM’s 2.68 mW — not a small number, but a structural zero.

It is small. One cell is essentially a via between two wires. An SRAM bit needs six transistors laid out side by side. This is where the $3.5\times$ density advantage in Chapter 10 comes from.

It is analog. This is the important one. The filament is not simply present or absent — you can grow it partially, by controlling how much current you allow during the SET pulse. A partly-grown filament gives an *intermediate* resistance. So one cell can store more than one bit.

Three states, not two

We use three resistance states per cell:

$$R_{\text{LRS}} = 542.8 \Omega, \quad R_{\text{MID}} = 1099.8 \Omega, \quad R_{\text{HRS}} = 218,587 \Omega$$

The ratio $R_{\text{HRS}}/R_{\text{LRS}} = 403$ is called the **on/off ratio**, and it is a headline figure of merit for a ReRAM device. Ours is good; many published devices sit near 10.

Why stop at three? Because ReRAM reliability degrades sharply beyond three or four levels — the filament is a physical object of a few atoms, and asking it to land reliably on eight distinguishable sizes is beyond what the device does repeatably. Three is what the physics comfortably supports, and (as Section 2.4 shows) three is exactly what our number format

needs. That coincidence is not luck; it is why the format was chosen.

1.3 Conductance is the natural language

Throughout this document we talk about **conductance** G rather than resistance R . They are reciprocals:

$$G = \frac{1}{R}$$

measured in siemens (S). The reason is Ohm's law. Written with resistance it is a division, $I = V/R$; written with conductance it is a multiplication,

$$I = V \cdot G$$

and multiplication is what a neural network needs. This single line is the entire reason analog in-memory computing exists.

Table 1.1: Our three states in both languages.

State	Resistance R	Conductance G	Meaning
LRS	542.8Ω	$1843 \mu\text{S}$	strong connection, “big weight”
MID	1099.8Ω	$909 \mu\text{S}$	half connection, “half weight”
HRS	$218,587 \Omega$	$4.6 \mu\text{S}$	almost nothing, “zero weight”

Notice that HRS is not exactly zero conductance — it is $4.6 \mu\text{S}$, small but real. That residual leakage matters later (Section 17.2), because it means our stored numbers do not land exactly on the values we intended.

Chapter 2

How a Grid of Cells Multiplies

2.1 One cell is one multiplication

Put a voltage V across a cell of conductance G , and the current that flows is $I = V \cdot G$. If we agree that the voltage represents an *input number* and the conductance represents a *weight*, then the current is their product. The chip did a multiplication, and it took no clock cycle, no multiplier circuit, and essentially no energy. Physics did it.

2.2 A column of cells is a dot product

Now stack several cells so that their bottom ends all connect to the same wire. Each cell i has its own input voltage V_i and its own conductance G_i . Each contributes a current $V_i G_i$ into the shared wire, and **Kirchhoff's current law** says currents joining at a node add up:

$$I_{\text{total}} = \sum_i V_i G_i$$

That is a **dot product** — multiply pairs, add the results. It is the single operation that dominates every neural network. And here it happens in one step, in the wire, for free.

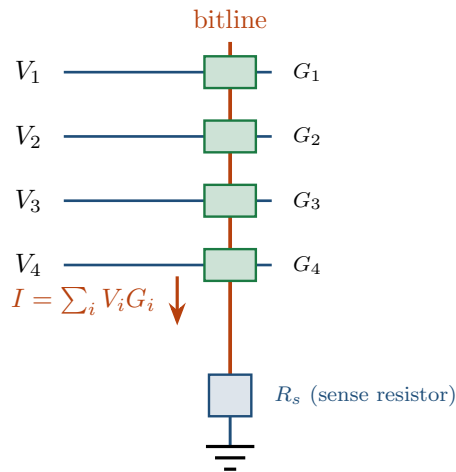


Figure 2.1: A single crossbar column. Four inputs, four weights, one dot product, computed by Kirchhoff's current law in the vertical wire. The current is read by measuring the voltage across a small *sense resistor* R_s at the bottom. That sense resistor is innocuous-looking and is the source of the entire problem this project solves.

2.3 A grid of cells is a matrix multiplication

Add more columns. Every column sees the same input voltages but has its own set of conductances, so every column computes a different dot product — against the same input vector. That is exactly a matrix–vector product:

$$\mathbf{I} = \mathbf{G}^T \mathbf{V}$$

A grid of M rows by K columns performs $M \times K$ multiply-accumulate operations in a single electrical settling time, which is nanoseconds. This structure is called a **crossbar**, and a crossbar

of a given size is called a **tile**. The number of rows M is the **tile height**, and it is the parameter this entire document is about.

Why tile height is the parameter that matters

At the bottom of every column sits an **ADC** (analog-to-digital converter) that turns the analog current into a number the digital part of the chip can use. ADCs are expensive — in our design they consume roughly 55% of both the energy and the area of a tile.

You need one ADC per column regardless of how many rows the column has. So a tile with 128 rows does four times as much work per ADC conversion as a tile with 32 rows. **Taller tiles amortise the ADC cost over more computation, and that is essentially the only lever available for improving the energy efficiency of a crossbar accelerator.** Everything that follows is a fight to make the tile taller.

2.4 The problem of negative numbers

Conductance cannot be negative. A resistor cannot push current backwards. But neural network weights are negative about half the time, so we need a representation for them.

The solution is **2T2R** — two transistors, two resistors — meaning each weight gets *two* cells on *two* bitlines. One bitline accumulates the positive parts, the other the negative parts, and we subtract the two currents at the end:

$$w_i \rightarrow (G_i^p, G_i^n), \quad w_i \propto G_i^p - G_i^n$$

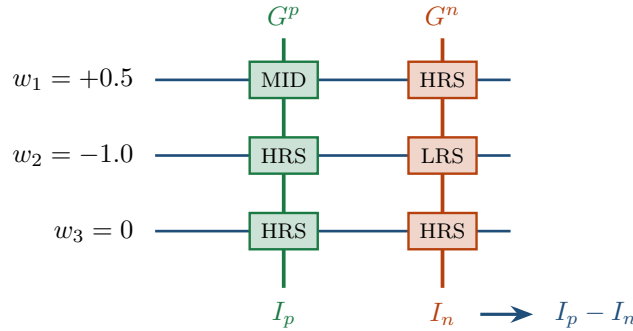


Figure 2.2: 2T2R differential encoding. A positive weight puts a low resistance on the + line and HRS on the − line; a negative weight does the reverse; a zero weight puts HRS on both. Subtracting the two column currents recovers a signed result. Crucially for this project, **the two bitlines are read separately**, which is what makes the fix in Chapter 6 possible.

Table 2.1: Complete weight-to-cell mapping. Three resistance states are enough to represent all five signed levels of our format.

Weight level	G^p cell	G^n cell	Resulting current
+1.0	LRS	HRS	large positive
+0.5	MID	HRS	small positive
0.0	HRS	HRS	≈ 0
−0.5	HRS	MID	small negative
−1.0	HRS	LRS	large negative

This is the coincidence promised earlier: five signed levels require only three distinct *magnitudes*, and three states is exactly what a ReRAM cell holds reliably.

Part II

The Number Format, and the Collision

Chapter 3

FP2: Two Bits per Weight

3.1 Why weights need to be small

A ResNet-18 network has about 11.16 million weights. Stored as standard 32-bit floats that is 44.6 MB, which does not fit on a chip. Every practical accelerator therefore **quantizes** — stores weights with fewer bits.

The extreme end of that spectrum is two bits per weight, a $16\times$ reduction. The obvious way to spend two bits is four evenly spaced integers. **FP2-E1M0** spends them differently.

3.2 The format

FP2-E1M0 means: 1 exponent bit, 0 mantissa bits, plus a sign. Each weight is one of five levels:

$$\{-1, -0.5, 0, +0.5, +1\}$$

Two bits give four codes, and five levels need slightly more than that — the format packs a group of weights together to recover the difference. The important point is not the bit-packing but the *spacing*: the levels are not evenly spaced. They cluster near zero.

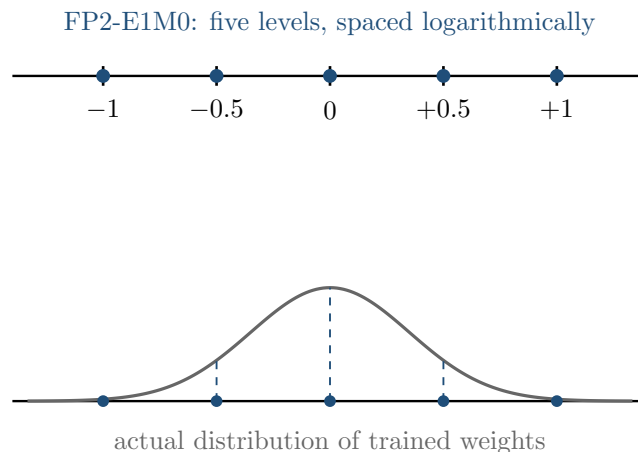


Figure 3.1: Why unequal spacing wins. Trained neural-network weights are bell-shaped — most are near zero, few are large. Placing levels densely near zero and sparsely at the extremes puts the precision where the weights actually are. This is the same argument that makes floating point better than fixed point.

3.3 Block floating point: the shared scale

Five fixed levels between -1 and $+1$ cannot represent a weight of 0.003 or of 47 . The format solves this by **block floating point**: a group of B consecutive weights shares one 8-bit scale factor s , and the actual weight is

$$w_{\text{actual}} = s \times \ell, \quad \ell \in \{-1, -0.5, 0, 0.5, 1\}$$

Each block picks its own s — typically the largest magnitude in the block — so a block of tiny weights and a block of large weights both get full use of the five levels.

The block size B

The FP2 specification fixes $B = 32$. Thirty-two consecutive weights share one scale factor. Storage cost: 32×2 bits for the levels, plus 8 bits for the shared scale, giving 2.25 bits per weight. The scale is nearly free.

Remember the number 32. The entire problem in Chapter 5 comes from it.

3.4 Quantization-aware training

Simply rounding a trained network’s weights to five levels — called **post-training quantization** or PTQ — loses accuracy, because the network was never told this would happen.

Quantization-aware training (QAT) retraines the network with the rounding simulated in the forward pass, so it learns weights that survive it. The trick required is the **straight-through estimator**: rounding has zero gradient everywhere, which would stop learning entirely, so during the backward pass we pretend the rounding was the identity function and pass the gradient through unchanged. Full-precision master weights are kept alongside and are what actually get updated.

Table 3.1: PTQ versus QAT on CIFAR-10, ResNet-18, full 10,000-image test set.

Configuration	Top-1 accuracy	Change vs FP32
FP32 baseline	92.40%	—
FP2, post-training quantization	87.88%	−4.52
FP2, 10 epochs of QAT	92.44%	+0.04

How to say this

“Two bits per weight sounds like it should be devastating. With quantization-aware training it costs us nothing measurable on CIFAR-10 — we land four hundredths of a point *above* the FP32 baseline, which is within run-to-run noise. On the harder CIFAR-100 task it does cost about 1.25 points, and we report that rather than only quoting the favourable result.”

Chapter 4

The Collision: Why Block Size *Is* Tile Height

4.1 Two numbers that must be equal

We now have two independent-looking parameters:

- B , the block size of the number format — how many weights share a scale factor. FP2 fixes this at 32.
- M , the tile height of the crossbar — how many rows feed one column. The hardware wants this large, 128 or more, to amortise the ADC.

They look unrelated. They are not. **They are forced to be the same number**, and understanding why is the key insight of the whole project.

4.2 The argument

A crossbar column physically produces *one* current. That current is the sum over every row in the column. To convert it back into a real number you multiply by the block's shared scale factor s .

Now suppose a column spans two different blocks, with scales s_1 and s_2 . The current arriving at the ADC is

$$I = \underbrace{\sum_{i \in \text{block 1}} V_i G_i}_{\text{needs } \times s_1} + \underbrace{\sum_{i \in \text{block 2}} V_i G_i}_{\text{needs } \times s_2}$$

The two halves have already been added together by physics, in the wire, before anything could be scaled. There is no way to apply two different multipliers to one number after the fact. The information needed to separate them is gone.

$$B = M$$

The block-floating-point block size and the crossbar tile height are not two parameters. They are one physical parameter with two names.

FP2 says it must be 32. The crossbar says it should be 128 or more. **They cannot both be satisfied.** That is the collision.

4.3 What happens if you build it anyway

The natural response is: fine, ignore FP2's $B = 32$ and just build a taller tile with a matching larger block. We did exactly that, and measured the accuracy of the real network running through the real circuit equations.

How to say this

“At the block size FP2 specifies, the array already costs us fifteen accuracy points. At the tile height that would actually make the hardware efficient, the network outputs random guesses. So the format and the circuit are in direct conflict, and this is the thing we set out

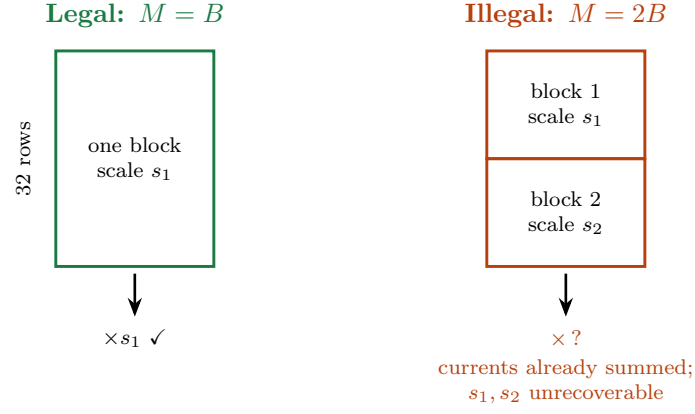


Figure 4.1: Why a tile cannot span two blocks. Once physics has added the currents together, no downstream multiplier can undo the mixing.

Table 4.1: The collapse. CIFAR-10 Top-1, full 10,000-image test set. “FP2 digital” is the same quantized network evaluated in exact arithmetic; “crossbar raw” is the same network on the physical array.

$B = M$	FP2 digital	Crossbar raw	Cost of the analog array
32	92.42%	77.66%	−14.76
64	92.47%	57.54%	−34.93
128	92.81%	11.60%	−81.21
256	92.43%	10.00%	−82.43

to explain.”

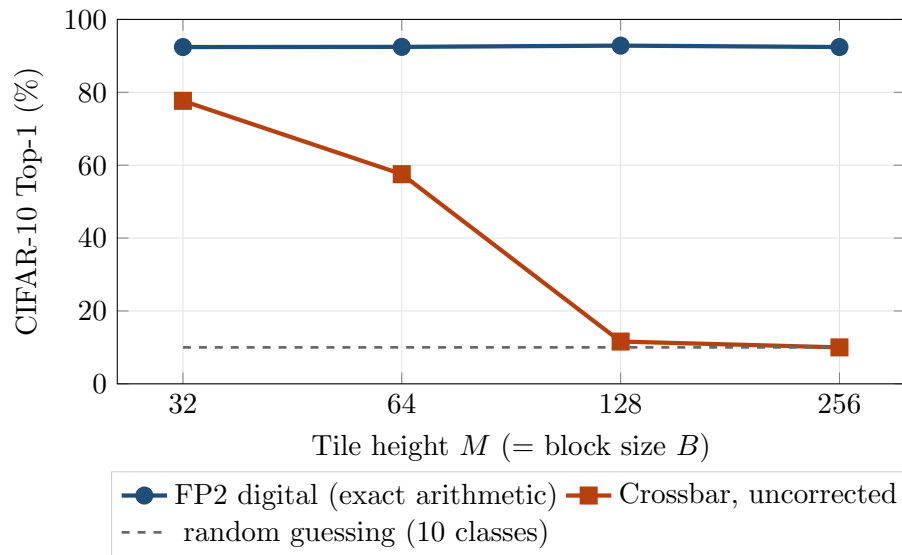


Figure 4.2: The central problem. Digital accuracy is flat — the block size does not bother the *network* at all. But on the physical array, accuracy falls off a cliff and reaches pure chance by $M = 128$. At the tile height the hardware wants, the chip has stopped computing anything.

Part III

The Cause and the Fix

Chapter 5

Diagnosing the Collapse

5.1 The suspect: the sense resistor

Look again at the column diagram. The current leaving the column has to be *measured*, and the standard way is to pass it through a small known resistor R_s and measure the voltage that develops across it. Ours is $20\,\Omega$ — deliberately small, so as to disturb the circuit as little as possible.

But $20\,\Omega$ is not zero. The moment current flows, the bottom of the column lifts off ground by IR_s . And the voltage across each cell is the difference between its input voltage and the bitline voltage. So lifting the bitline *reduces the voltage across every cell simultaneously*, which reduces every cell's current, which is not what the ideal equation said.

5.2 Deriving the error

Let the column have cells $G_1 \dots G_M$ driven at voltages $V_1 \dots V_M$, all feeding a bitline held at unknown voltage v_b , which drains to ground through R_s . Kirchhoff's current law at the bitline node says everything flowing in must flow out:

$$\sum_i (V_i - v_b)G_i = \frac{v_b}{R_s}$$

Expand and collect the v_b terms:

$$\sum_i V_i G_i = v_b \left(\frac{1}{R_s} + \sum_i G_i \right)$$

which gives the bitline voltage in closed form:

$$\boxed{v_b = \frac{\sum_i V_i G_i}{\frac{1}{R_s} + G_{\text{col}}}}, \quad G_{\text{col}} \equiv \sum_i G_i$$

The measured current is $I_{\text{actual}} = v_b/R_s$, and the ideal answer we wanted was $I_{\text{ideal}} = \sum_i V_i G_i$. Dividing one by the other:

$$\boxed{I_{\text{actual}} = \frac{I_{\text{ideal}}}{1 + R_s G_{\text{col}}}}$$

Read that boxed equation carefully

The output is the correct answer divided by $(1 + R_s G_{\text{col}})$.

Now ask: what is in that denominator? Only R_s , which is a fixed component value, and G_{col} , the sum of the conductances programmed into that column.

The input voltages do not appear. The error does not depend on the data at all. It depends only on what we programmed into the array — which we chose, and therefore know exactly, at compile time.

This is not noise. It is a *fixed, known, calculable gain*.

5.3 Why the collapse gets worse with taller tiles

G_{col} is a sum over M cells. Double the tile height and you roughly double it, which makes the denominator larger and the attenuation worse. That is the exact shape of the collapse in Chapter 4.

Table 5.1: Fraction of the ideal current that actually survives to the ADC.

Tile height M	32	128	256
Current retained	73.9%	41.5%	26.0%
Signal lost	26.1%	58.5%	74.0%

At $M = 256$ three-quarters of the signal is gone before the ADC sees it. The network is not confused — it is being shouted at through a wall.

5.4 Why the literature missed this

A methodological point worth making in a viva

The standard way to evaluate an analog CIM design is to model the analog error as additive zero-mean Gaussian noise injected into the layer outputs. Nearly every published simulator does this.

A zero-mean additive model **structurally cannot represent a systematic gain**. It assumes the error averages to nothing over many terms. Our error is a 26–74% *attenuation*, which never averages away.

An evaluation built on that assumption would have observed the accuracy collapse, attributed it to noise accumulating in tall columns, concluded the tile-height limit was fundamental, and stopped. We only saw the truth because we solved the actual circuit equations. **Finding the fix required throwing out the standard methodology**, and that is itself a contribution.

Chapter 6

The Fix: Per-Column Gain Calibration

6.1 The idea

If the output is the right answer multiplied by a known constant, multiply by the reciprocal.

For column j , compute once, at compile time:

$$c_j = 1 + R_s \sum_i G_{ij}$$

and after reading the column, multiply the result by c_j .

That is the entire fix. One number per column. A multiplication that folds into the block-scale multiply already present in the digital path, so it costs no extra hardware at all.

6.2 Doing it correctly with 2T2R

There is a subtlety worth getting right, because the naive version does not work.

The positive and negative bitlines have *different* total conductances — they hold different cells — so they suffer *different* attenuation factors. If you subtract first and correct afterwards, you are applying one correction to a mixture of two differently-corrupted quantities, and it does not cancel.

The order must be: read each bitline separately, correct each with its own constant, *then* subtract.

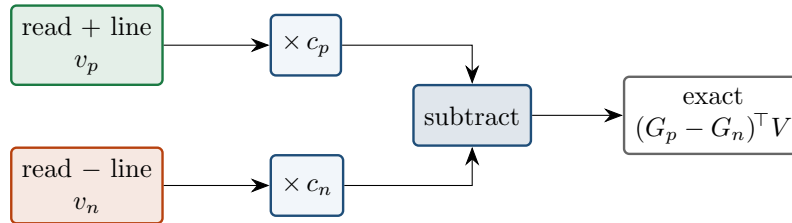


Figure 6.1: Correct order of operations. Each bitline is corrected by its own constant *before* the differential subtraction. Correcting after subtracting does not cancel, because the two lines have different total conductances.

6.3 The algebra collapses

Write it out. Each line, corrected:

$$c_p \cdot \frac{(G_p^\top V)}{c_p} = G_p^\top V, \quad c_n \cdot \frac{(G_n^\top V)}{c_n} = G_n^\top V$$

and subtracting,

$$(G_p - G_n)^\top V$$

which is precisely the ideal matrix–vector product we wanted in the first place. Not approximately. Exactly.

6.4 It works

Table 6.1: The headline result. CIFAR-10 Top-1, full 10,000-image test set.

$B = M$	Crossbar raw	Calibrated	Recovered	FP2 digital ceiling
32	77.66%	92.42%	+14.76	92.42%
64	57.54%	92.47%	+34.93	92.47%
128	11.60%	92.81%	+81.21	92.81%
256	10.00%	92.43%	+82.43	92.43%

The calibrated column equals the digital ceiling at every tile height, to the last digit. There is no residual error to trade off, because the correction is algebraically exact rather than a fit.

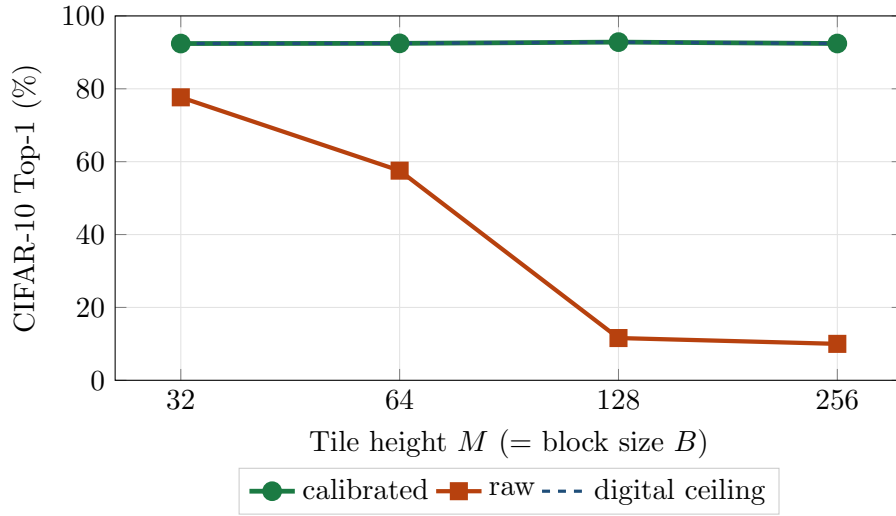


Figure 6.2: The fix in one picture. The calibrated curve sits exactly on the digital ceiling — the dashed line is hidden underneath it — while the uncorrected curve falls to chance. Tile height has gone from a binding constraint to a free design parameter.

How to say this

“The output is the correct answer divided by one plus R_s times the column conductance. That denominator contains no activation term — it depends only on the weights, which we programmed ourselves. So it is a constant we can compute at compile time and simply divide back out. Because 2T2R already reads both bitlines separately, correcting each one before the subtraction makes the algebra collapse to the exact ideal product. One constant per column, folded into a multiply that already exists in the datapath, and the accuracy comes back completely.”

6.5 An unexpected consequence: the sense resistor becomes free

There is a second payoff, and it took an outside paper to notice it.

Roy, Patil and Shanbhag (ISCAS 2022) worked out analytically what limits the accuracy of any resistive crossbar. They found that the sensing resistance has an *optimum*. Too small and the signal is weak, so the ADC’s quantisation steps eat it. Too large and the signal clips against the converter’s range. The two effects balance at one value, and for ReRAM they put that value at $835\,\Omega$.

Our design used $20\ \Omega$ — forty times smaller. Not by analysis, but by necessity: the loading error $1 + R_s G_{\text{col}}$ grows with R_s , so an uncalibrated array is pushed toward the smallest sense resistor it can tolerate, and pays for that with a weak signal.

Calibration removes the loading term exactly. So the reason for staying small disappears. That is a testable prediction, and it was tested.

Table 6.2: Sensing resistance sweep, $B=M=128$, 6-bit ADC, full 10 000-image test set. Digital ceiling 92.81%.

$R_s\ (\Omega)$	Raw	Calibrated	Gap
20	11.61%	92.43%	−0.38
100	10.00%	92.62%	−0.19
200	10.00%	92.59%	−0.22
400	10.00%	92.58%	−0.23
600	10.00%	92.52%	−0.29
835	10.00%	92.59%	−0.22
1200	10.00%	92.30%	−0.51

Read the raw column first. From $100\ \Omega$ upward it is pinned at exactly 10.00% — one class in ten, pure chance. An uncalibrated crossbar *cannot operate at the analytically optimal sense resistance at all*. It is not merely worse there; it is destroyed.

Now the calibrated column: 92.30 to 92.62% across a sixty-fold range of R_s . A spread of 0.32 points. At the optimal $835\ \Omega$ it is within 0.22 points of the digital ceiling — and slightly *better* than at the $20\ \Omega$ we had been using.

How to say this

“An independent group derived the SNR-optimal sense resistance for ReRAM crossbars analytically and got 835 ohms. Our uncalibrated array is at chance there — it cannot use that operating point. With calibration we are within 0.22 points of digital, and accuracy is flat within a third of a point across a sixty-fold range of sense resistance. So the correction does not just recover accuracy at one operating point; it hands back the sense resistor as a free design parameter, including the value that independent analysis says is optimal.”

6.6 The same paper explains the device sweep

That paper gives a second result which lines up with something we had measured but could not explain.

They prove that compute SNR depends on the resistive *contrast* $R_{\text{off}}/R_{\text{on}}$ and *not* on the absolute value of R_{on} , because raising R_{on} scales signal and noise equally. They also find the benefit of higher contrast saturates once it passes roughly 12 to 15; beyond that, DAC mismatch and cell variation dominate instead.

Both match our measurements, which were taken before we read the paper:

- We swept R_{LRS} over $184\times$ with the contrast held at 403, and calibrated accuracy stayed within 0.40 points. Their theory says it should — SNR does not depend on absolute R_{on} .
- We tested an MRAM-like device at a contrast of 3 and got chance. Their threshold is 12–15, so a contrast of 3 is far below the point where the device stops being the limiter. It should fail, and it does.

An experiment agreeing with an independent analytical prediction is worth much more than either result alone. Say so.

Part IV

The Evidence

Chapter 7

Proving the Simulator Is Right

Everything above rests on our circuit model being correct. If the model were wrong, the fix would be a fix to a fiction. So this was verified exhaustively.

7.1 The three levels

Table 7.1: Validation chain. Each level is checked against the one above it.

Level	Checked against	Worst disagreement
ngspice DC solve	— (reference)	—
Closed-form nodal model	ngspice, 87,168 tiles	$3.41 \times 10^{-7}\%$
Vectorised GPU solver	nodal model	$< 10^{-12}$ relative
End-to-end CNN inference	vectorised solver	identical code path

ngspice is the standard open-source circuit simulator — the same class of tool used to sign off real silicon. We did not spot-check. We pushed **every tile of every layer** of the network through it: 87,168 tiles, 357 million individual MAC reads, 22 minutes on 15 CPU cores, zero errors.

The worst disagreement anywhere was 3.41×10^{-7} percent. That is roughly the precision of the arithmetic itself.

7.2 Why the fast model is fast

The closed-form solution from Chapter 5,

$$v_b = \frac{\sum_i V_i G_i}{1/R_s + \sum_i G_i}$$

requires no iteration — unlike a general circuit simulator, which must solve a system numerically. Because it is a single expression, it vectorises into one matrix multiplication, and all tiles of a layer batch into a single GPU call.

That is the difference between evaluating 10,000 test images in minutes versus weeks, and it is what made every full-test-set number in this document affordable.

Chapter 8

Where Every Number Comes From

A reviewer’s first question about any simulation study is which numbers were measured and which were assumed. This chapter answers that in one place.

8.1 Three tiers, printed on every result

Every number the tools emit carries one of three tags.

Table 8.1: Epistemic tiers. These are printed alongside the results, not buried in a footnote, because conflating them is how simulation studies quietly overstate their case.

Tag	Meaning	Trust
[SIM]	measured by ngspice on the netlist	high
[MODEL]	closed form derived from device parameters	medium
[ASSUM]	literature-typical placeholder, no PDK	low — replace before publishing

The honest one-line summary

The array is simulated. The periphery is modelled. The array power term is measured by ngspice for all 19 layers. Area, delay, ADC and DAC energy are closed-form models resting on literature-typical constants.

Awkwardly, the array is only 1.3% of tile area, so the part unique to this work is the well-measured part, and the part that dominates the headline efficiency figure is the assumed part. That is worth saying out loud before someone says it for you.

8.2 Where the device numbers came from

The three resistance states were obtained by sweeping RESET pulse amplitude and width against the `rram_v1.0.0` Verilog-A compact model, reading each back with a small non-disturbing read pulse, and selecting operating points that land in clearly separated bands.

The single largest caveat in the whole project

These are derived from a *compact device model*, not measured on fabricated silicon. Everything downstream inherits that. If the real device differs, the numbers move — though the *structure* of the argument (that the loading error is a deterministic function of programmed conductance) does not depend on the particular values.

8.3 The constants that stopped being assumptions

When this chapter was first written, most of the periphery constants carried [ASSUM]. Reading the source literature closed nearly all of them. The table below is the current state, because a reviewer will ask and the answer should be one lookup rather than an argument.

Table 8.2: Periphery constants and their sources. “Derived” means the number was computed here from what the paper reports rather than quoted from it; the arithmetic is recorded in CITATIONS.md so it can be checked.

Constant	Source
ADC 20 fJ/conv-step	Firlej <i>et al.</i> , JINST 2024. 10-bit SAR, 65 nm, 50–90 MSps. Derived FoM 15.9–24.4 fJ; our value sits between their 40 and 80 MSps points, at matching node and architecture.
1T1R cell 40 F^2	Bracketed by two fabricated parts: 31 F^2 at 90 nm (Shim <i>et al.</i> , SSC-L 2020) and 84 F^2 at 40 nm.
6T SRAM cell 140 F^2	Measured range is 150–200 F^2 (same Shim paper). Our value is <i>below</i> it, i.e. generous to SRAM.
SRAM leakage 120 pW/bit	Verma & Chandrakasan, JSSC 2008 — 65 nm, no node scaling. 8.39 pW/bit derived at 0.35 V; their own DIBL figure ($\geq 10\times$ from 0.3 V to 1 V) puts 1.2 V at 125–170 pW/bit.
SRAM read 0.31 pJ/bit	Horowitz ISSCC 2014, 32 KB cache, 20 pJ per 64-bit access.
INT8 MAC 0.59 pJ	Horowitz Fig. 1.1.9: 0.2 + 0.03 pJ at 45 nm 0.9 V, scaled $\times 2.57$ to 65 nm 1.2 V.
DRAM 10 pJ/bit	Horowitz, improved-I/O floor. His contemporary figure is > 20 ; the optimistic one is used so the baseline is not strawmanned.
Write ~ 100 pJ/cell	Perez <i>et al.</i> , TED 2021. One 1.2 V, 1 μ s pulse at 10–40 μ A is 12–48 pJ; write-verify applies several.
MAC unit area	Still uncited. Excluded from every reported area figure. It is 1.2% of the digital baseline’s footprint, where the buffer is 99%, so nothing depends on it.

Two of these corrections went against us, and that is the point

The SRAM read energy was originally 0.05 pJ/bit and the FP2 MAC 0.03 pJ. Both were roughly an order of magnitude too kind to the digital baseline we were comparing against — which is to say, both flattered our result.

Fixing them changed the conclusion. Under the old constants the digital baseline actually *beat* in-memory compute at high weight reuse (66.5 against 65.6 TOPS/W). Under the cited constants it does not, at any reuse. The result we now report is both stronger and better founded, and it was found by checking the numbers that helped us rather than only the ones that hurt.

8.4 The assumptions that matter most, and their sensitivity

Rather than defend a single value for the uncertain constants, each was swept and the spread reported.

Table 8.3: Headline efficiency against the ADC figure of merit alone. Everything else is held fixed.

ADC source	pJ/MAC	TOPS/W	die area
placeholder (FoM 20 fJ)	0.0246	81.3	412 mm ²
NeuroSim-sourced area	0.0246	81.3	329 mm ²
published SAR (FoM 9.5 fJ)	0.0141	142.2	364 mm ²

A 1.75× swing from one assumed constant

The headline efficiency figure is dominated by the ADC figure of merit, which is the least-grounded number in the model. Reporting the spread is more defensible than picking the flattering value — and it pre-empts the reviewer who asks where 20 fJ came from.

8.5 What the energy model does not count

The hardware model counts array, ADC, DAC and partial-sum accumulation. It has *no category* for interconnect, buffers, pooling or activation units. NeuroSim, which does account for those, puts them at 20.8% of per-image energy.

How to say this

“Our 81.3 TOPS/W is a tile-level figure — it counts the array, converters and accumulation. Adding NeuroSim’s 20.8% share for interconnect and buffers would put the comparable chip-level number near 64 TOPS/W. Most of the apparent advantage over NeuroSim’s 29.3 is an accounting difference rather than a design difference, and we say so.”

8.6 A 30× error, caught and fixed

The fallback array-power model was once 30× too high — 2587 μW per tile modelled against 85.8 μW measured. It assumed every cell sat at nominal conductance with the full read voltage across it. In 2T2R only the magnitude side of a nonzero weight leaves HRS, half of those sit at the intermediate state, and post-ReLU activations are sparse.

The error was *conservative* — every corrected efficiency figure came out better. The fallback is now anchored to the ngspice measurement and reproduces 85.8 μW exactly at the reference point.

Chapter 9

Does It Survive Real Devices?

A calibration computed from *nominal* conductances is only useful if the actual manufactured cells are close to nominal. They are not — real ReRAM cells scatter substantially.

9.1 Device-to-device variability

We model programming scatter as log-normal, drawn once per device and then frozen (a manufactured chip does not re-roll its defects). We then calibrate using the *nominal* values — deliberately pretending we never measured the real chip. This is called **blind calibration**, and it is the cheap option: no per-cell read-back, no per-chip characterisation.

Table 9.1: Blind calibration under device variability. Ten independent device instances per point, full test set. σ is the conductance spread.

$B = M$	$\sigma = 0$	$\sigma = 10\%$	$\sigma = 20\%$	Loss at 20%
32	92.42	92.32	91.88	−0.54
64	92.47	92.34	91.86	−0.61
128	92.81	92.66	92.03	−0.78
256	92.43	92.25	91.62	−0.81

Worst case, at 20% device spread and the tallest tile, is 0.81 points. We also compared against write-verify (measuring each cell after programming and calibrating from the measurement): it performs the same. **The expensive option buys nothing**, which is a useful practical result.

A retraction, kept deliberately visible

An earlier draft claimed that *taller columns are more variability tolerant*. That came from a run using 1,000 test images and 3 random seeds, which showed −1.27 points at $B = 32$ against +0.07 at $B = 256$.

On the full 10,000-image test set with 10 seeds, the ordering **reverses**: taller columns are slightly *worse*, monotonically, as the table shows.

The original claim was sampling noise. At $n = 1000$ the 95% confidence interval is roughly ± 1.9 points, which cannot support a claim about differences of a few tenths of a point. The claim is withdrawn. What survives — and what we actually assert — is that the cost is under one point everywhere.

If asked in a viva why this is in the document: because an examiner who finds it themselves is a much worse outcome than one who sees you found it first.

9.2 Conductance drift: the result that changed the design

ReRAM cells do not hold their conductance forever. The filament slowly relaxes and conductance decays as a power law,

$$G(t) = G_0 \left(\frac{t}{t_{\text{ref}}} \right)^{-\nu}$$

with a small exponent ν that differs between states and varies from cell to cell.

Table 9.2: Drift. “Calibrated once” means constants computed at manufacture and never updated. “Refreshed” means recomputed from the drifted array.

B	Elapsed time	Raw	Calibrated once	Refreshed
32	1 second	79.06	92.31	92.15
32	1 hour	46.23	88.81	91.98
32	1 day	31.45	84.22	91.92
32	1 month	19.05	74.94	91.89
32	1 year	13.97	65.17	91.87
128	1 year	10.00	72.21	92.04
256	1 year	10.00	80.56	91.73

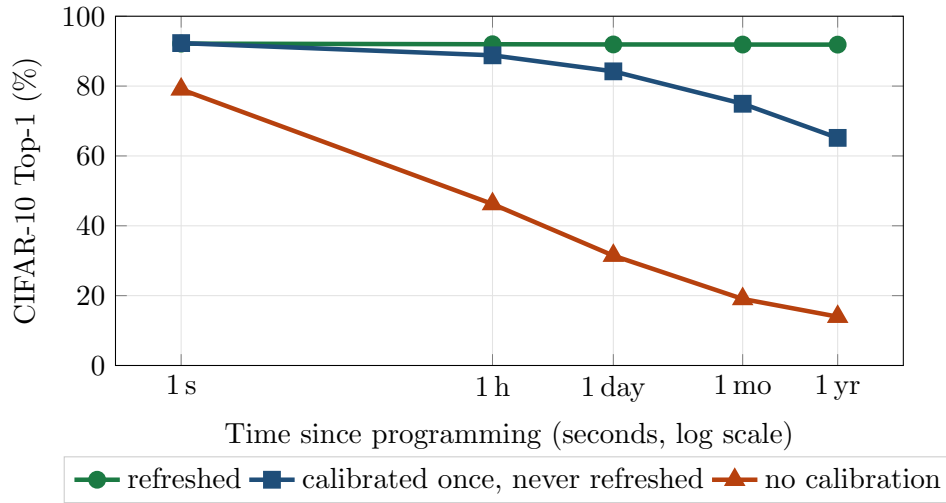


Figure 9.1: Drift at $B = 32$. A calibration computed once at manufacture decays badly — 27 points lost after a year. Recomputing it from the drifted array holds accuracy flat indefinitely.

This matters enormously for our claim, because our calibration constants were computed from conductances that are no longer accurate.

This forced an honest weakening of the claim

We would like to say “compile-time calibration”. Drift says we cannot. The honest claim is **calibration with periodic refresh**.

The saving grace is that drift is a power law, so damage accumulates with the *logarithm* of time. Refreshing at 1 minute, 1 hour, 1 day, 1 week, 1 month and 1 year — roughly **ten refreshes over the entire product lifetime** — bounds the error to well under a point. A refresh reads the array and recomputes one constant per column. That is cheap and it folds into the block-scale multiply already in the datapath.

We also got the refresh mechanism wrong at first. Recomputing only the loading constant left 15.7% residual error — *worse* than leaving it stale. The reason: drift shrinks the weights themselves, and that scale error dominates the loading error. The fix was to add a least-squares per-column gain folded into the block scale, which corrects both at once.

9.3 How much do the drift assumptions matter?

The drift conclusion rests on σ_ν , the cell-to-cell spread in the drift exponent, which is the least certain input in the entire model. Rather than assert one value, it was swept.

Table 9.3: Drift-exponent spread sweep, $B=128$, one year elapsed. “Refresh worth” is the accuracy regained by recomputing the constants.

σ_ν	stale	refreshed	refresh worth
0.001	71.52	92.12	+20.60
0.002	71.53	92.03	+20.50
0.004	71.73	91.73	+20.00
0.008	73.36	90.54	+17.18
0.016	77.91	80.29	+ 2.38

The boundary of the refresh claim

Refresh holds within 2 points of the ceiling up to $\sigma_\nu = 0.004$ and **begins failing at 0.008**. By 0.016 refreshing is nearly worthless.

The operating assumption of 0.004 sits right at the edge of the safe region. The honest statement is therefore: *“periodic refresh restores accuracy to within one point provided the cell-to-cell drift-exponent spread stays below roughly 0.008; beyond that, drift rather than the divider limits tile height.”* Stating the boundary is far stronger than asserting the point.

9.4 Confirming the mechanism rather than just observing it

An earlier observation — that stale-calibration loss falls as tile height grows — was made *after* looking at the data, which is the weakest kind of claim. The proposed explanation, that the error scales with $R_s \cdot G_{\text{col}}$, makes a falsifiable prediction: the tile-height trend must strengthen as R_s grows and wash out as it shrinks.

Table 9.4: Stale-calibration cost (points below the ceiling after one year with no refresh) against sense resistance and tile height. Less negative is better.

R_s (Ω)	$B=32$	$B=64$	$B=128$	$B=256$	spread
5	−31.56	−28.32	−35.16	−26.83	−4.73
10	−29.47	−25.44	−29.43	−20.57	−8.90
20	−25.69	−20.56	−21.08	−12.88	−12.81
40	−19.66	−13.67	−12.18	−6.38	−13.28
80	−12.03	−7.46	−5.54	−2.89	−9.14

Stale cost shrinks monotonically with R_s at every tile height. The ordering across tile heights is clean and monotonic at $R_s \geq 20 \Omega$ but scrambled at $R_s = 5 \Omega$, exactly where the loading term is too small to dominate.

Why this matters more than the numbers

“The trend was found by looking at data, so we treated it as a hypothesis rather than a result. The mechanism predicts it should scale with R_s . We swept R_s over sixteen-fold and the prediction held. That converts an observation into an explanation, and it answers the objection that the trend was cherry-picked after the fact.”

Chapter 10

How Many ADC Bits Do We Need?

The ADC dominates energy and area, and **SAR** ADC energy scales as 2^N for N bits. Every bit removed halves the cost; going from 8 bits to 6 divides the ADC energy by four. So this question is worth real money.

Table 10.1: ADC resolution sweep, gain-calibrated, full test set. “Exact” means an infinitely precise ADC.

B	4 bits	5 bits	6 bits	8 bits	Exact	Required
32	74.88	91.27	92.41	92.47	92.42	6
64	66.93	90.97	92.27	92.47	92.47	6
128	77.18	91.43	92.43	92.70	92.81	6
256	77.88	90.85	92.01	92.49	92.43	6

The structural finding: the requirement is flat in B

Six bits suffice at *every* tile height. An eight-times taller tile needs no extra ADC resolution. This matters more than it looks. The worry with tall tiles is that they produce a wider range of output values and therefore demand a finer ADC — which would hand back exactly the efficiency you gained by amortising the ADC over more rows. **It does not happen.** The saving is real and it is kept.

A second correction, same cause as the first

An earlier draft claimed *five* bits sufficed. That was measured at 1000–2000 images. On the full test set, five bits costs 1.15–1.58 points at every B . The requirement is six. Both retractions in this document have the same root cause: a number that looked settled at small sample size. The rule adopted afterwards — re-run anything you intend to quote, on the full test set — is worth stating as a lesson learned.

Chapter 11

What the Model Leaves Out

The calibration corrects one thing exactly: the shared sense resistor. Real interconnect has other parasitics, and honesty requires measuring them rather than hoping they are small.

11.1 Solving the whole mesh

The closed-form model treats the wordline as an ideal voltage source and the bitline as a single node. A separate solver drops both assumptions and solves the full two-dimensional resistive mesh — every wordline segment, every bitline segment, every crosspoint — as one sparse linear system.

Its self-test asserts that with ideal wires it reproduces the closed-form model to 8.8×10^{-8} relative. That agreement is what makes any *difference* at nonzero wire resistance trustworthy as physics rather than a solver bug.

Table 11.1: Each parasitic isolated, $M=128$. The sanity row must be near zero after calibration; anything far above it is an effect the per-column constant does not model.

Case	raw error	after calibration
ideal wires (sanity check)	71.3%	0.0002%
wordline metal $0.5\ \Omega$	71.8%	1.82%
wordline driver $50\ \Omega$	78.3%	25.4%
bitline metal $0.5\ \Omega$	84.8%	49.1%
bitline metal $1.0\ \Omega$	89.1%	64.0%

11.2 The uncomfortable finding

Wordline *metal* — the parasitic everyone worries about — is mild. The two that hurt are **driver output resistance** and **bitline metal resistance**.

The reason is structural. The correction $1 + R_s G_{\text{col}}$ treats the bitline as a single lumped node. Real bitline resistance is *distributed*: a cell in row 0 and a cell in row 127 see different accumulated IR on the way to the sense node. No single per-column scalar can represent that.

Does a fitted constant rescue it? No.

A per-column gain was fitted by least squares on one set of activation patterns and tested on *unseen* ones — the only version of the test that means anything, since a gain fitted and evaluated on the same data would look perfect regardless.

M	raw	calibrated	+ fitted gain (held out)
32	57.7%	35.9%	14.1%
64	70.1%	40.6%	17.7%
128	87.8%	55.9%	47.4%
256	91.6%	62.8%	68.7%

At $M=32$ a fitted constant recovers most of it. At $M=256$ it makes things *worse* — the fit does not transfer to unseen activations at all. And even the best case, 14.1%, is five orders of magnitude away from the 0.0002% the sense-resistor calibration achieves.

11.3 What this does to the claim

The claim, narrowed to what the evidence supports

Before: “the analog error is a compile-time constant.”

After: “the shared-sense-resistor loading term is exactly correctable at compile time. Distributed wire parasitics are not, and are reported separately.”

That is a smaller claim. It is still a real one — the term being corrected is the one that causes the accuracy collapse in the first place — and it is a claim that survives contact with a mesh solver, which the larger version does not.

These runs use assumed wire resistances on random tiles, not the trained network with a real metal pitch. The right next step is a PDK metal pitch, which would say whether R_{bl} is nearer 0.1 or $1.0\ \Omega$ — the difference between a 17% and a 64% residual.

Chapter 12

Where ReRAM Actually Wins

An honest evaluation has to say where the technology does *not* help. ReRAM’s advantage is not raw compute throughput. It is memory.

12.1 Density, standby power, and data movement

Table 12.1: ReRAM against 6T SRAM for the same 11.16 M weights at 65 nm.

Metric	ReRAM	SRAM	Advantage
Weight array area (2-bit weights)	3.77 mm ²	13.20 mm ²	3.5×
vs. FP32 weights in SRAM	3.77 mm ²	211.2 mm ²	56×
Standby power	0 mW	2.68 mW	structural
Weight-fetch energy (reuse = 1)	0 pJ/MAC	0.100 pJ/MAC	structural

Two of these are infinite rather than large, and the reason is structural rather than a matter of degree. A non-volatile cell needs no power to remember. A weight that is physically part of the computation needs never to be fetched.

That last point deserves emphasis. In a conventional accelerator, moving a weight from memory into the multiplier costs far more energy than the multiplication itself. In a crossbar the weight *is* the multiplier, so that cost is not reduced — it does not exist.

12.2 The efficiency result

Table 12.2: Baseline versus proposed design. The baseline is not a straw man — it is what you get by building FP2 and a conventional passive-readout crossbar exactly as each is specified.

	Baseline	Proposed
Block size / tile height $B = M$	32	128
Per-column gain calibration	no	yes
ADC resolution	8 bit	6 bit
CIFAR-10 Top-1	77.66%	92.43%
Gap to FP2-digital ceiling	−15.15 pts	−0.38 pts
Energy per MAC	0.323 pJ	0.0237 pJ
Energy efficiency	6.2 TOPS/W	81.3 TOPS/W

Why both improve at once

Normally accuracy and efficiency trade against each other. Here they move together, because a single change unlocks both.

Removing the loading error makes tall tiles *usable*. Tall tiles amortise the ADC over four times as many rows. And because the ADC requirement is flat in B , the 8-bit converter can also drop to 6 bits, dividing its energy by four again.

+14.77 accuracy points and 13.1× energy efficiency, simultaneously.

12.3 Writing is brutally expensive

Reprogramming a ReRAM cell costs roughly 100 pJ. Reading one costs about 0.001 pJ — five orders of magnitude less.

The consequence is architectural. If the network does not fit on the chip all at once, and you swap weights in and out, you pay that write cost constantly. Our model puts the **break-even batch at about 2,200 images**: below that, the chip spends more energy writing weights than computing with them and is better described as a write engine than an accelerator.

This is a real limitation, and it should be stated plainly

The design only makes sense *fully weight-stationary* — every weight gets its own permanent cells. At $B = 128$ our model reports 412 mm² of die area for ResNet-18 fully resident, which is not manufacturable at 65 nm.

The honest framing: this is a *scaled-node* argument. The area shrinks with the square of feature size, so what is impossible at 65 nm is comfortable at 7 nm. LLaMA-7B at FP2 needs ~2265 mm² at 65 nm but only ~26 mm² at 7 nm.

12.4 Independent cross-check against NeuroSim

NeuroSim V1.4 is a widely used, independently developed CIM simulator. Running a comparable configuration:

Table 12.3: Cross-check against NeuroSim V1.4.

	This work	NeuroSim
ADC share of energy	55.6%	68.0%
ADC share of area	55.2% [†]	19.2%
Energy efficiency	13.8 TOPS/W	30.3 TOPS/W

[†] **Different denominators, and the row is only honest with this footnote.** Our 55.2% is of *tile* area — array, converters, input drive and accumulation. NeuroSim’s 19.2% is of *chip* area, which also contains interconnect, global buffer and pooling. The numbers are not in conflict and must not be read as a disagreement.

Different tool, different methodology, same qualitative conclusion: the ADC is a leading cost. NeuroSim’s area figure comes from transistor-level sizing and is better grounded than ours, so we adopt it.

12.5 The chip area breakdown, and a lesson about searching

For a long time this project believed that seventy per cent of NeuroSim’s reported chip area was unaccounted for — that the printed components did not add up to the printed total, and that an area argument therefore could not be made. That belief was wrong, and how it survived is worth recording.

The components were being extracted from the log with a text search that required the word “area” in the line. NeuroSim prints this:

```
Other Peripheries (e.g. decoders, mux, switchmatrix, buffers, IC, pooling and activation
units) on chip : 1.15e+07um^2
```

That line contains neither “area” nor “array”. It was silently dropped, and a dropped component looks exactly like missing silicon. Two other lines were missed the same way. **A search**

that fails and a quantity that is genuinely absent produce identical output, and only one of them is a finding.

The fix was to match on the *unit* instead — any line of the form `label : number um^2` — so the label can be anything. With that, the breakdown closes.

Table 12.4: NeuroSim V1.4 chip area, VGG-8, 65 nm. Total 80.06 mm².

Component	Area (μm ²)	Share
CIM array	21,031,200	26.3%
Accumulation circuits	16,177,500	20.2%
ADC	15,384,400	19.2%
Interconnect	12,141,100	15.2%
Other peripherals	11,549,600	14.4%
Named subtotal	76,283,800	95.3%
Residual	3,772,300	4.7%

The 4.7% residual is not mysterious either. Reading `ChipCalculateArea` in `Chip.cpp`, nine statements add to the area accumulator, and several of them — the global H-tree or bus, the global buffer, pooling, global accumulation, ReLU and sigmoid — have no corresponding print. Name it “global buffer, interconnect and activation units” and move on.

What this changes

The ADC is not the largest area term. At 19.2% it sits third, behind the array at 26.3% and accumulation at 20.2%. Our own tile-level model puts the ADC at 55%, and both can be true: the tile-level figure excludes interconnect, global buffer and pooling, so it is a percentage of a much smaller denominator. **Always state which denominator is being used.** Putting 55% and 19.2% in the same table without that qualification is the kind of thing a reviewer catches.

An area *improvement* can now be argued properly, because the breakdown closes: run the baseline and proposed configurations and difference them component by component, with `neurosim_area_breakdown.py --log A --log B`. Attribute any saving to the component that moved. If the total falls while the array stays fixed, the saving came from the converters, and that is a different claim from a density claim.

Two caveats on this comparison

One. NeuroSim runs VGG-8; we run ResNet-18. This is not like-for-like and must be labelled as such.

Two. The device conversion is only half applied. NeuroSim’s C++ core carries our device parameters correctly, but its Python wrapper has a *separate* `--onoffratio` parameter that was still at its default of 10 against our device’s 403. The comparison must be re-run before it is quoted in a paper.

Chapter 13

Does It Transfer to Transformers?

CNNs are one workload. A reviewer will ask whether the result is specific to them. So we tested a LLaMA-architecture language model.

13.1 What can and cannot go on a crossbar

Table 13.1: Transformer layer mapping.

Operation	Mappable?	Why
q, k, v, o projections	yes	static weights
gate, up, down (FFN)	yes	static weights
$QK^\top$ (attention scores)	no	activation $\times$ activation
AV (weighted values)	no	activation $\times$ activation

A crossbar stores something fixed. Attention multiplies two things that both change every token, so there is nothing to program into a cell that stays put, and reprogramming per token is hopeless at 100 pJ per cell. **The attention core stays digital.** This is a genuine architectural limitation and the paper must say so.

13.2 The negative result, reported honestly

Applying FP2 post-training quantization to LLaMA-160M destroys it: perplexity goes from 28.02 to 445,867.

The diagnosis is clean. **69.5% of weights round to zero.** LLM weight distributions are far more heavy-tailed than a CNN's — each block's scale is set by a single outlier, and everything else in the block is then small enough relative to that scale that it rounds away. A simple Gaussian calculation predicts about 52%; the real distribution gives 69.5%.

This is the quantizer's failure, not the crossbar's

It is essential to keep these separate. The perplexity collapse is caused by 2-bit quantization of a heavy-tailed weight distribution without retraining. It has nothing to do with the crossbar. Conflating the two would be the first thing a reviewer catches.

13.3 Isolating the crossbar claim

Since perplexity cannot separate the two effects, we measured the thing our contribution is actually about: **readout error**. We hooked all 84 target linear layers, captured the real activations the model produces, and pushed them through the crossbar three ways — ideal, raw, and calibrated.

134 dB is float32 machine precision. The loading error is not reduced — it is *gone*.

How to say this

“Six orders of magnitude, across 84 layers of a real transformer driven by real activations. The heavy-tailed weight distribution that wrecks 2-bit quantization has no effect at all on the

Table 13.2: Per-layer readout error on real LLaMA-160M activations, $B = 128$, 6-bit ADC, mean over 84 layers.

	Raw	Calibrated
Mean relative readout error	27.7%	$\mathbf{2.08 \times 10^{-5}\%}$
Signal-to-noise ratio	~ 11 dB	$\sim \mathbf{134}$ dB

calibration, and the reason is structural: the correction depends only on column conductance sums, never on how the weights are distributed. So the contribution transfers from CNNs to transformers unchanged — and we can show that even though the quantized language model itself is unusable without retraining.”

Part V

Defending the Work

What Is Genuinely New

14.1 The claimed contributions

1. **Per-column loading-gain calibration turns crossbar tile height from a binding constraint into a free parameter.** Exact rather than approximate, one constant per column, folded into a multiply that already exists, requiring no per-cell read-back.
2. **Blind calibration survives realistic device variability** — under one point at $\sigma = 20\%$, computed from nominal conductances. Write-verify buys nothing, which is a useful practical result.
3. **$B = M$: block-floating-point block size and CIM tile height are one physical parameter.** We believe this is the first joint accuracy / SNR / energy characterisation of that coupling.
4. **Methodological: additive-Gaussian analog error models cannot express a systematic gain**, so work built on them structurally cannot find this failure mode. This is the first circuit-exact end-to-end evaluation validated against exhaustive SPICE across every tile of every layer.
5. **QAT makes block size a hardware-only parameter** — with retraining, accuracy is flat in B , so the block size can be chosen purely for the circuit’s benefit.

14.2 The most likely challenge, and the answer

“Compensating crossbar loading is not new.”

Correct, and you should concede it immediately rather than defend it. IR-drop and loading compensation in crossbars is an established area.

The narrower claims that survive:

- The correction is *exact*, not a fitted approximation, and we show the algebra collapsing to the ideal product.
- It requires *no read-back* — blind, from nominal values, and we measure that this is sufficient under 20% device spread.
- It is tied specifically to *block floating point*, where it removes a constraint that the number format itself imposes. The $B = M$ framing is, as far as we know, new.
- The 2T2R ordering matters and is not obvious: correcting each bitline before subtraction is what makes it exact.

A literature search on “IR drop compensation ReRAM crossbar”, “conductance-dependent output scaling CIM”, and “sneak-path calibration in-memory computing” should be completed before the introduction is written, because it decides whether the paper says “first to” or “unlike prior work”.

Chapter 15

Standing Against Prior Work

Compensating crossbar non-idealities is a crowded field. This chapter names the competing approaches and states honestly which parts of the contribution survive contact with them.

15.1 The landscape

Table 15.1: Existing approaches to crossbar non-ideality. The two in bold are direct threats and must be addressed in the introduction, not left for a reviewer to raise.

Approach		Level	Cost
Weight distortion (non-ideality-aware training)	pre-	algorithmic	requires retraining
BatchNorm calibration	re-	software	retraining-free; needs validation data
Op-amp virtual ground (TIA)	virtual	circuit	one op-amp per column
Peripheral injection	current	mixed-signal	op-amps, current mirrors
Sparsification		algorithmic	accuracy/sparsity trade

15.2 Threat one: why not just use a transimpedance amplifier?

The most dangerous question in the viva

A transimpedance amplifier holds the bitline at *true* virtual ground. With $v_{bl} = 0$ the term $1 + R_s G_{col}$ never arises at all.

So the error being corrected is a consequence of choosing passive resistive sensing. Left unaddressed, the entire work reads as an elaborate solution to a self-inflicted problem.

The answer is not that the circuit fix is impossible — it is that it is not free, and the cost can be priced on the same tile.

The number that settles the argument

The static power entry. **384 μW per tile is 4.5 $\times$ what the array itself dissipates**, it is always on whether or not the chip is computing, and across the network it adds **2.10 W** to a design whose entire compute budget is 274 nJ per inference.

The digital correction costs one multiply per column, folded into a multiply that already exists.

Three consequences, in order of how much they matter:

- **Static power.** An op-amp burns bias current continuously. This is the entry that decides it.

Table 15.2: What per-bitline transimpedance sensing would cost, $M=128$, $K=16$. One amplifier per sensed bitline, and two converters per column are already required by the calibration, so 32 amplifiers. Amplifier area and bias are placeholders for a compact two-stage OTA at 65 nm — $1000\text{ }\mu\text{m}^2$ and $10\text{ }\mu\text{A}$ at 1.2 V — and are labelled [ASSUM] accordingly.

	passive + calibration	TIA sensing
extra analog blocks	none	32 op-amps/tile
tile area	$75\,369\text{ }\mu\text{m}^2$	$107\,369\text{ }\mu\text{m}^2$ (+42%)
die area, whole net	411.8 mm^2	586.6 mm^2
added static power	0	$384\text{ }\mu\text{W/tile}$ (2.10 W total)
vs. measured array power	—	$4.5\times$
correction cost	1 multiply/col	none needed

- **Area.** 42% more tile area, pushing an already-marginal 411.8 mm^2 die to 586.6 mm^2 .
- **Bandwidth.** The passive path settles in about 1 ps against a 12 ns ADC conversion, so sensing is currently free in the latency budget. An amplifier would not be.

Be honest about what these numbers are

The amplifier area and bias current are [ASSUM] placeholders, not PDK values. The *direction* of the conclusion is robust — an amplifier per column cannot be free, and static power cannot be zero — but the exact multiplier would move with a real op-amp design. Present it as an order-of-magnitude argument, which is all it needs to be.

15.3 Threat two: BatchNorm recalibration

BatchNorm recalibration is retraining-free, absorbs systematic column-current attenuation, and is reported in the literature as landing within 0.16% of floating point. That is close to this work’s claim by a much simpler route.

Four differences are real:

1. **Exact, not fitted.** The correction here is an algebraic identity that collapses to the ideal product. BatchNorm recalibration is a statistical fit and leaves a residual.
2. **Per-column, not per-layer.** BatchNorm parameters are per output channel and absorb a layer-average attenuation. The loading gain differs *column to column* because G_{col} differs, and that spread grows with tile height.
3. **No data required.** BatchNorm recalibration needs a validation set; this correction is computed from the programmed conductances alone.
4. **Works without BatchNorm.** The transformer layers measured in Chapter 11 have none, so a BatchNorm-based fix is simply unavailable there.

A prediction that failed, and was dropped

The argument above was that BatchNorm recalibration should degrade as tile height grows, because a per-channel scalar cannot capture per-column variation in G_{col} , and that variation widens with M .

Table 15.3: BatchNorm recalibration measured against per-column calibration. Full 10 000-image test set, 6-bit ADC. BatchNorm statistics fitted on train-split images.

$B=M$	ideal	raw	BN	per-col	BN gap	per-col gap
32	92.42	78.90	90.75	92.35	−1.67	−0.07
64	92.47	57.47	90.09	92.09	−2.38	−0.38
128	92.81	11.53	90.62	92.42	−2.19	−0.39
256	92.43	10.00	90.24	91.88	−2.19	−0.55

It was tested and it did not hold. The BatchNorm gap averages 2.02 points below $B=128$ and 2.19 above it — flat within noise. The argument has been removed from the paper rather than quietly softened.

What the measurement gives instead, which is stronger

- **Per-column wins at every tile height.** Worst gap 0.55 points against BatchNorm’s 2.38, a factor of 4.3. At $B=32$ it is 0.07 against 1.67, a factor of 24. A result that holds everywhere is cleaner than a trend that holds only at large M .
- **BatchNorm recalibration is a strong baseline, and saying so helps.** It lifts $B=256$ from chance (10.00%) to 90.24%. That a statistical fit absorbs most of the error is itself evidence the error is systematic rather than random.
- **Exactness, needing no calibration data, and working without BatchNorm all stand untouched.** Those were always the stronger three arguments.

The detail to volunteer before it is asked

“Our own gap grows slightly with tile height, from 0.07 to 0.55 points. That is not calibration residual — the correction is exact. It is 6-bit ADC quantization, whose relative cost grows as the partial-sum range widens with taller tiles. The ADC sweep shows the same effect independently.”

One discrepancy to report rather than hide

Prior work reports BatchNorm recalibration landing within 0.16% of floating point. Here it lands 1.67–2.38 points short.

The likely reason is that their attenuation is IR-drop induced and roughly uniform across a layer, whereas the shared-sense-resistor gain varies with each column’s programmed conductance — so a per-channel affine has less to grip on. The discrepancy should be stated, not resolved by quietly quoting whichever number flatters.

15.4 What to claim, and what to concede

Concede immediately

“First to compensate crossbar loading” is **false**. IR-drop and loading compensation is an established field. Conceding this in the first paragraph costs nothing and buys credibility for the narrower claims that *are* defensible.

The claim that leads

The strongest contribution is not the calibration — it is the **format–device match** and the $B=M$ coupling it forces.

Multi-level ReRAM reliably holds about *three* states. FP2’s five signed levels need only *three magnitudes*, because 2T2R supplies the sign. So FP2 lands exactly on what the device can physically hold: INT8 would need 256 levels or eight cells per weight; binary discards the dynamic range block floating point exists to provide.

FP2 was designed for digital packing efficiency, with no reference to resistive memory. That it coincides with the three-state limit of a ReRAM cell is a genuine co-design observation — and once FP2 is the format, $B=32$ is fixed, and $B=M$ forces that onto the crossbar. Every later step then follows from a physical constraint rather than a design preference.

The whole argument in the order it should be told

1. FP2 is what a three-state ReRAM cell can hold.
2. So the format is fixed, and with it $B = 32$.
3. $B = M$, so the crossbar tile height is fixed too.
4. Which forces a tile too short to amortise the ADC.
5. The loading gain is what makes taller tiles fail.
6. And it is exactly correctable, for one multiply per column.

Chapter 16

Anticipated Questions

Why is the error a gain rather than noise? Because the derivation gives $I_{\text{actual}} = I_{\text{ideal}}/(1 + R_s G_{\text{col}})$, and the denominator contains only the sense resistance and the programmed conductances. No activation term appears. Anything that does not depend on the data is not noise.

Why must block size equal tile height? A column produces one current, and that current must be multiplied by one scale factor. If the column spanned two blocks, two different scales would have to be applied to a sum that physics already combined. The information needed to separate them is gone.

Why three resistance states? Two reasons that happen to agree. ReRAM reliability degrades sharply beyond three or four levels. And FP2’s five signed levels need only three *magnitudes*, because 2T2R supplies the sign. INT8 would need 256 states, or eight cells per weight.

Why not just use a smaller sense resistor? It reduces the error but does not remove it, and the measured voltage IR_s shrinks with it — so you trade a gain error for a signal too small to digitise. Calibration removes the error exactly without touching the signal.

How do you know your simulator is right? 87,168 tiles — every tile of every layer — through ngspice, 357 million MAC reads, zero errors, worst disagreement 3.41×10^{-7} percent.

Why do accuracy and efficiency improve together? One change unlocks both. Removing the loading error makes tall tiles usable; tall tiles amortise the ADC over more rows; and because the ADC bit requirement is flat in B , the converter also drops from 8 bits to 6, whose energy scales as 2^N .

What is the weakest part of the work? Three things, in order. *Drift* forces the claim down from “compile-time” to “with periodic refresh”. *Die area* at 65 nm is 412 mm² fully resident, so the argument depends on scaled nodes. *Wordline IR drop* is not yet modelled, and unlike bitline loading it is activation-dependent, so it would not be a compile-time constant — this is the most likely thing to weaken the contribution.

Why did the LLaMA perplexity fail? Because 2-bit post-training quantization of a heavy-tailed weight distribution rounds 69.5% of weights to zero. That is the quantizer, not the crossbar. The crossbar’s own contribution is measured separately as readout error, where calibration reduces 27.7% to $2 \times 10^{-5}\%$.

Why not use a transimpedance amplifier? It would work, and it would remove the error entirely. It also costs an op-amp per column in area and static power, and its bandwidth would bound the read cycle where the passive path currently settles in 1 ps against a 12 ns conversion. The argument is that loading need not be fixed in the circuit, not that it cannot be.

How is BatchNorm recalibration different from this? It is a per-layer statistical fit needing validation data; this is a per-column algebraic identity needing none. It is also unavailable on networks without BatchNorm, such as the transformer layers measured here.

Is 81.3 TOPS/W a chip-level number? No, and it should never be quoted as one. It counts array, converters and accumulation. Adding interconnect and buffers at NeuroSim’s 20.8% share

puts the chip-level figure near 64 TOPS/W.

How sensitive is the result to the ADC assumption? Very. The figure of merit alone swings the headline between 81.3 and 142.2 TOPS/W, which is why all three variants are reported rather than one.

What would you do next? Model wordline IR drop; complete the novelty search; run ImageNet; and apply QAT to the language model so the transformer result can be stated as perplexity rather than only as readout error.

Chapter 17

Honest Limitations and Remaining Work

17.1 What is not yet done

Table 17.1: Remaining work, ordered by whether it blocks submission.

Priority	Item	Note
Blocking	Novelty search	Decides whether the introduction claims priority.
Blocking	Re-run Pareto at 6-bit	Efficiency figures were computed at 5 bits and rescaled by hand.
Blocking	Regenerate figures 1–7	They still carry $n = 2000$ numbers.
Important	Error bars	Ten seeds exist; tables currently print means only.
Important	Sweep drift exponent spread	The refresh conclusion rests on $\sigma_\nu = 0.004$, the least certain input.
Important	Level-mapping error	See below.
Important	NeuroSim re-run	At the correct on/off ratio, and VGG-8 vs ResNet-18 must be labelled.
Strengthening	Wordline IR drop	The most likely remaining threat.
Strengthening	ImageNet	Strongest single credibility upgrade.
Strengthening	Read noise	Averages over a dot product, so likely benign — worth showing.
Strengthening	Temperature	Both drift and conductance are strongly temperature dependent.
Deferred	RTL / synthesis flow	Blocked: the Verilog top level has no behavioural ADC model, which is why RTL error has read N/A throughout.
Deferred	Tapeout-grade area	Needs a real PDK; everything is currently an assumption.

17.2 The level-mapping error

Recall that HRS is not infinite resistance. Working through the actual device numbers, the three programmed states do not land exactly on $\{0, 0.5, 1\}$:

$$\frac{R_{\text{MID}}}{R_{\text{LRS}}} = \frac{1099.8}{542.8} = 2.0262 \quad (\text{should be exactly } 2)$$

and finite HRS leakage subtracts a further 0.25% from every level. The net effect is that level 0.5 is physically 0.491 — a 1.79% error.

This is *separate* from the readout error this project fixes, and it is currently absorbed into what we call the digital ceiling. It is also straightforwardly fixable: either trim R_{MID} at programming time, or fold the measured ratio into the block scale factor. It should be quantified and then either fixed or reported.

17.3 Lessons that generalise

The pattern behind every mistake made

Five errors were caught and corrected during this work:

- “Taller columns are more variability tolerant” — retracted; it was sampling noise at $n = 1000$.
- “Five ADC bits suffice” — wrong for the same reason; six are needed.
- The hardware model’s fallback array power was $30\times$ too high, making every efficiency figure pessimistic.
- A figure caption was hardcoded and kept claiming “5 bits” after the answer had changed to 6.
- Refreshing only the loading constant made drift *worse*, because weight shrinkage dominates.

Every one is either a number that looked settled at small sample size, or a caption that stopped tracking its data. The rule that follows: **re-run anything you intend to quote, on the full test set, and derive captions from the data rather than typing them.**

Chapter 18

Summary

The complete argument in nine steps

1. A ReRAM cell stores a number as a conductance. Ohm's law $I = V \cdot G$ makes reading it a multiplication.
2. A grid of cells performs a full matrix–vector product in one electrical settling time.
3. Each column needs an ADC, which dominates energy and area, so taller tiles are the main efficiency lever.
4. FP2 stores weights as five levels with a scale shared across 32 weights. Three resistance states plus 2T2R differential encoding express all five.
5. A column produces one current and can therefore carry only one scale factor, so block size and tile height are the same parameter: $B = M$. FP2 wants 32; the circuit wants 128+.
6. Built naively, accuracy collapses from 92% to 10% — pure chance.
7. The cause is $I_{\text{actual}} = I_{\text{ideal}} / (1 + R_s G_{\text{col}})$, which depends only on programmed conductances. It is a known gain, not noise, and Gaussian error models cannot represent it.
8. Correcting each bitline by its own constant before the differential subtraction collapses algebraically to the exact ideal product. Accuracy returns fully at every tile height.
9. Tile height becomes free, the ADC drops to 6 bits, and the result is +14.77 accuracy points and $13.1\times$ energy efficiency at once — verified against exhaustive SPICE, surviving 20% device variability, and requiring periodic refresh to survive drift.

If you have sixty seconds

“We store 2-bit floating-point neural network weights directly in ReRAM cells, so multiplication happens by Ohm's law. The format shares a scale factor across 32 weights, and because a crossbar column produces a single current that can only carry one scale, the block size is forced to equal the tile height. That is a problem, because the circuit wants tall tiles to amortise the ADC. When we built it, accuracy fell to random guessing. We found the cause was not noise but a deterministic gain — the output is attenuated by one plus the sense resistance times the column conductance, which depends only on the weights we programmed. Dividing it back out, one constant per column, recovers the accuracy exactly. That frees the tile height, which lets us go four times taller and drop the ADC from eight bits to six, giving fourteen accuracy points and thirteen-fold energy efficiency simultaneously. It is verified against SPICE on every tile of every layer, it survives twenty percent device variability, and it transfers to transformers — though it needs periodic recalibration to survive conductance drift.”